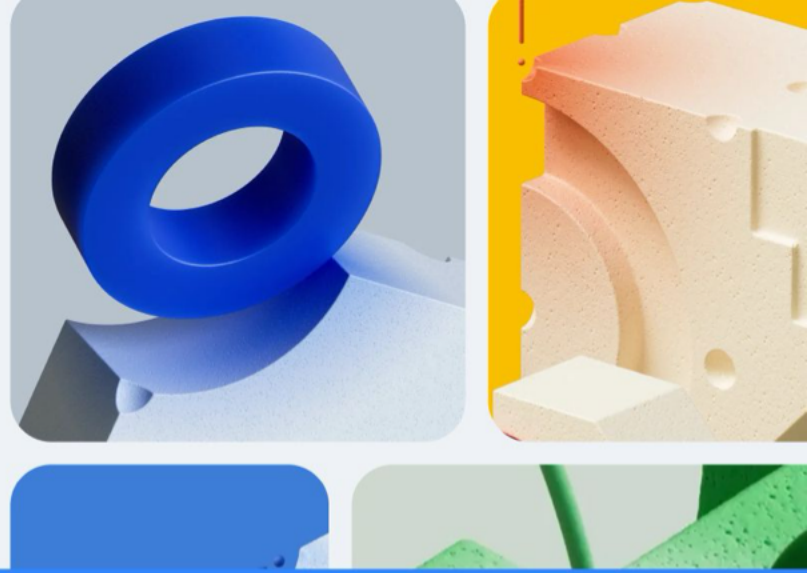
Google DeepMind | Google Cloud

Startup Buildathon

India 2026

Build your next big thing.

Powered by H2S | nasscom | HEADSTART



Dhalam AI — Real-Time Coercion & Sideloading

Team Ladydebuggers (2 Members) | Pitch Deck

Anusha Jain (Systems Architect, AI Engineer at IBM) & Abhyudaya Marya (Product Strategy, Genpact)

The Problem: India's ₹1,750+ Cr Social Engineering & Coercion Crisis

Scammers exploit psychological duress (fake Police 'Digital Arrest') to force victims to install legitimate remote-access utilities (AnyDesk) and drain accounts. Antivirus checks file signatures and fails; LLM chatbots are blind to device state. Panicked victims have no way to verify threats or device safety.

The Crisis of Digital Arrest & Social Engineering in India

A person under pressure needs a combined assessment of conversational context and device safety.

1 Psychological Duress

Fraudsters forge official CBI warrants or fake bill cutoffs to create panic.

2 Weaponized Utilities

Victims are directed to install remote support tools or grant Accessibility access.

3 The Systems Disconnect

Antivirus misses legitimate tools; chatbots analyze text in a vacuum without device context.

The user's question

Is this notice genuine?
Is my screen recorded?
How do I secure this?

Dhalam helps the person pause, correlate clues, and revoke dangerous access.

Problem framing: Social engineering and device-access hijacking; blind spots in traditional antivirus.

Dhalam AI — Interlocked, Closed-Loop Coercion

Consent before analysis. Explain the evidence. Correlate with telemetry. Guide remediation.

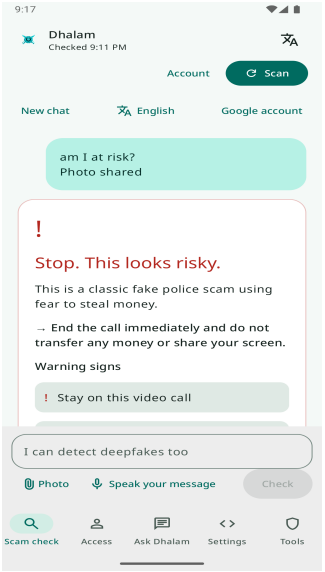
Proposed Solution Workflow

• 01 Multimodal Triage
Ingests screenshots & text. Gemini 3.8 Flash strips EXIF GPS & triages coercion risk.

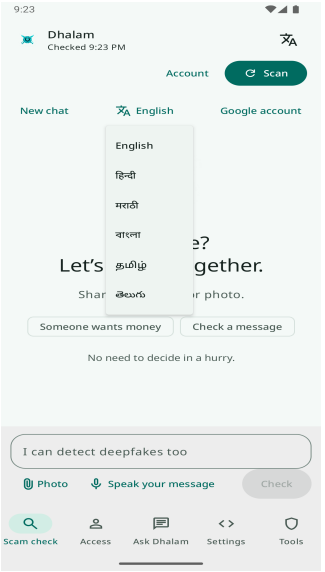
• 02 Correlate Telemetry
App audits active screen recording and enabled Accessibility tools in real time.

• 03 Grounded Advice
sqlite-vec retrieves official RBI circulars with strict citation checks.

• 04 Remediate & Verify
Settings deep-links launch deactivation; re-scan confirms device is secure.



Native Compose App



WhatsApp Triage Bot

Security boundaries: Privacy compliance, size-bounded payloads, zero raw screenshot persistence.

Systems Architecture: Interlocking AI & OS Telemetry

Clients expose only the device observations their operating system permits.

CLIENT FRONTENDS

Native Android Compose App
Meta WhatsApp Bot

Friction-free Credential Manager Google Sign-In, Live accessibility status, and screen capture detection.

SERVERLESS CLOUD

FastAPI in Mumbai (asia-south1)
Cryptographic Google ID token verification
Sliding-window rate limits

Pillow EXIF metadata sanitizer

AI & VECTOR

Vertex AI Gemini 3.8 Flash
Structured JSON schema
gemini-embedding-001 (768-dim)

In-process sqlite-vec vector DB with strict citation matching

Enterprise Security & Privacy

Uploaded images are stripped of GPS/EXIF metadata before AI dispatch. No screenshots are persisted on Cloud Run or Vertex AI. Size-bounded, consent-gated,

Native Android accesses AccessibilityManager capabilities and screen capture callbacks; iOS is restricted.

Google AI & Cloud Platform Core Integration

Complete architectural alignment across Google's modern developer ecosystem.

Gemini on Vertex AI

Structured multimodal scam assessments using strict Pydantic ModelAssessment schema enforcement.

gemini-embedding-001

Generates 768-dim embeddings powering in-process sqlite-vec vector RAG with $0 DB cluster cost.

Google Cloud Run

Hosted containerized API in Mumbai (asia-south1) for local data residency and ultra-low latency.

Android Credential Manager

Frictionless Google OAuth 2.0. Plus Google Safe Browsing API v4, and Google Noto Sans Indic typography.

Two Google AI Models, Two Specialized Roles

Structured multimodal reasoning and semantic vector indexing working in harmony.

Gemini (Vertex AI)

Input: Consented text or screenshots.

Output: Strict Pydantic JSON schema mapping coercion tactics, threat severity, and physical safety actions.

Reasoning: Evaluated with synthetic and user-consented examples across 6 Indic languages.

gemini-embedding-001

Role: Embeds queries and the official RBI regulatory advisory corpus for vector retrieval.

Moat: Powering the sqlite-vec index running in-process on Cloud Run with $0 database cost.

Grounding: Programmatic citation validation blocks LLM hallucinations.

AI acts as an intelligent assistant to human verification, not an automated safety arbiter or transaction blocker.

AI evaluation parameters: False reassurance, unsupported claims, multilingual clarity, and user comprehension.

Market Viability & Go-To-Market Business Model

High commercial value solving a critical digital banking liability for financial institutions.

TAM / SAM & Acquisition

TAM: ₹3,600 Cr ($430M) covering 350M+ UPI users in India. SAM: ₹750 Cr ($90M) vulnerable groups and RBI-regulated banks.

Consumer subscriptions are unviable; free app & WhatsApp bot drive organic brand trust and top-of-funnel.

B2B2C Embedded Banking SDK

Embedded SDK for Indian commercial banks (HDFC, SBI, ICICI) and UPI payment apps (PhonePe, Google Pay).

Moat: Telemetry checks run before authorizing transfers > ₹10,000, reducing bank fraud liability. SaaS license at $0.02 / MAU or micro-fee.

Execution Proof & Unit Economics

Gemini Flash check costs ~7 paise ($0.0008); in-process vector DB runs at zero cost, yielding >85% gross margins. Live deployment

Competitive differentiation: Interlocked context & telemetry correlation outperforms standalone scanners.

Google DeepMind | Google Cloud

Startup Buildathon

— India 2026

Build Your Next Big Thing.

Powered by H2S | nasscom | HEADSTART



Dhalam AI

Turn a pressured moment into a verified, protected decision.

Ladydebuggers • **Anusha Jain & Abhyudaya Marya**

Live Endpoint: https://dhalam-api-xxpghctsqq-el.a.run.app/health